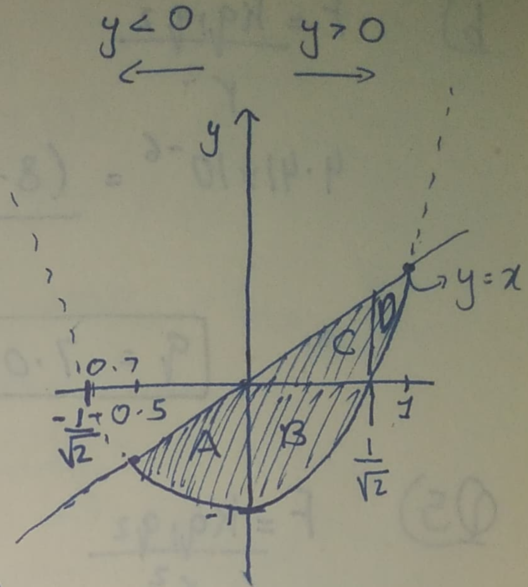
Maths Assignment 3



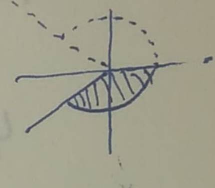
Q1) $y = x \;;\; y = 2x^2 - 1$

$2x^2 - 1 = 0$

$x = \sqrt{\frac{1}{2}}$

$x = \pm\frac{1}{\sqrt{2}}$

$$V_1 = \left[\pi \int_{-\frac{1}{\sqrt{2}}}^{+\frac{1}{\sqrt{2}}} (2x^2-1)^2\,dx\right] - \left[\pi \int_{-\frac{1}{\sqrt{2}}}^{0} (x)^2\,dx\right]$$



$$V_1 = \left[\pi \int_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} 4x^4 - 4x^2 + 1\,dx\right] - \left[\pi \int_{-\frac{1}{\sqrt{2}}}^{0} x^2\,dx\right]$$

$$V_1 = \left[\pi\left[\frac{4x^5}{5} - \frac{4x^3}{3} + x \Big|_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}}\right]\right] - \left[\pi\left[\frac{x^3}{3}\Big|_{-\frac{1}{\sqrt{2}}}^{0}\right]\right]$$

$$V_1 = \left[\pi\left(0.377 - (-0.377)\right)\right] - \left[\pi\left(0 - \left(-\frac{\sqrt{2}}{12}\right)\right)\right]$$

$V_1 =$ ~~1.999~~ 1.999 (~~a~~ Volume of Region A and B)

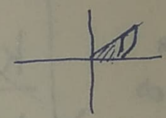
$$V_2 = \left[\pi \int_0^1 (x)^2\,dx\right] - \left[\pi \int_{\frac{1}{\sqrt{2}}}^{1} (2x^2-1)^2\,dx\right]$$

$$V_2 = \left[\pi \int_0^1 x^2\,dx\right] - \left[\pi \int_{\frac{1}{\sqrt{2}}}^{1} 4x^4 - 4x^2 + 1\,dx\right]$$

$$V_2 = \left[\pi \left[\frac{x^3}{3}\Big|_0^1\right]\right] - \left[\pi \left(\frac{4x^5}{5} - \frac{4x^3}{3} + x \Big|_{\frac{1}{\sqrt{2}}}^{1}\right)\right]$$

$$V_2 = \frac{\pi}{3} - 0.089$$

$$V_2 = 0.958$$



$$V_3 = \pi \int_0^{\frac{1}{\sqrt{2}}} \left(-(2x^2-1)\right)^2 dx$$

$$V_3 = \pi \int_0^{\frac{1}{\sqrt{2}}} 4x^4 - 4x^2 + 1\,dx$$

$$V_3 = \pi \left[\frac{4x^5}{5} - \frac{4x^3}{3} + x \Big|_0^{\frac{1}{\sqrt{2}}}\right]$$

$$V_3 = 1.184$$

$$V_4 = \pi \int_0^{0.5} (2x^2-1)^2 - (x)^2$$

$$= \pi(0.3167 - 0)$$

$$= 0.995$$

$$\therefore V_0 = 1.184 - 0.995 = 0.189$$

$$V_T = 1.999 + 0.189$$

$$\boxed{V_T = 2.19 \text{ units}^3}$$ Answer

Q2) Divergence test

$$\sum_{k=1}^{\infty} \frac{\sqrt{k}}{\sqrt{k^2+1}}$$

$$\lim_{k \to \infty} \sqrt{\frac{k}{k^2+1}}$$

$$\lim_{k \to \infty} \sqrt{\frac{\cancel{k^2}\left(\frac{1}{k}\right)}{\cancel{k^2}\left(1+\frac{1}{k^2}\right)}}$$

$$\lim_{k \to \infty} \sqrt{\frac{\left(\frac{1}{k}\right)}{\left(1+\frac{1}{k^2}\right)}}$$

$$= \sqrt{\frac{\left(\frac{1}{\infty}\right)}{\left(1+\frac{1}{\infty^2}\right)}}$$

$$= \frac{0}{1+0}$$

$$= \frac{0}{1}$$

$$= 0$$

$$\boxed{\text{As } \lim_{k \to \infty} = 0 \ \therefore \text{ the test yields no conclusion}}$$

Q3) $\sum_{k=0}^{\infty} -ke^{-k}$ $\quad f(x) = -xe^{-x}$

The integral test is not possible as series will never have any positive term. And for the test to occur the ~~seriest~~ series should have / must have positive terms.
(Pg 583 our calculus Book states [Summary of Convergence test table])

However if at any case it can be done, than I suppose it will be done like this,

$$\lim_{b\to\infty} \int_0^b f(x)\,dx$$

integration by parts

$$\lim_{b\to\infty} \int_0^b -xe^{-x}\,dx$$

$$-\int_0^b xe^{-x}\,dx$$

$$uv - \int v\,du$$

$$= -\left[-xe^{-x} - \int -e^{-x}\,dx\right]$$

$$= -\left[-xe^{-x} + \int e^{-x}\,dx\right]$$

$$= -\left[-xe^{-x} + \int e^{u}\,du\right]$$

$$= -\left[-xe^{-x} - e^{-x}\right]$$

$$= xe^{-x} + e^{-x} \Big|_0^b$$

Substitute $u = -x$

$du = -dx$

$$\int e^u\,du = \frac{e^u}{\ln(e)} (-)dx$$

$$= -e^u$$

$$= e^{-x}(x+1)\Big|_0^{\infty}$$

$$= \left(e^{-\infty}(\infty+1)\right) - \left(e^{-0}(0+1)\right)$$

$$= \infty(\infty) - 1$$

$$= \infty - 1$$

$$= \infty$$

As answer is infinity it is diverging